Operations Research 1: Deterministic Models
Linear Programming: Duality

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Example: Product Mix (Wyndor Glass Co.)

Wyndor Glass Co.

- Two products
  - Product 1
    - Glass door w/ aluminum framing
  - Product 2
    - Wood-frame window
- Three plants
  - Plant 1
    - Aluminum frames and hardware
  - Plant 2
    - Wood frames
  - Plant 3
    - Glass and product assemble
- Data
  - Available production time in each plant (hours / week)
  - Production time used in each plant for each batch of each product (hours / batch)
  - Profit per batch of each product

	Doors	Windows		
Profit per batch	$3,000	$5,000		
				Hours
	Hours used per batch produced			available
Plant 1	1	0		4
Plant 2	0	2		12
Plant 3	3	2		18

- Decision: production
- Objective: to maximize the total profit (per week)

Wyndor Glass Co. (Revisit)

TOpS

- Mathematical formulation
  - x_1 = # batches of product 1 produced per week
  - x_2 = # batches of product 2 produced per week
  - z = total profit per week

	Doors	Windows			
Profit per batch	$3,000	$5,000			
					Hours
	Hours used per batch produced				available
Plant 1	1	0			4
Plant 2	0	2			12
Plant 3	3	2			18
	Doors	Windows			Total profit
Batches produced					

$$
\begin{aligned}
\max \quad & z = 3x_1 + 5x_2 \\
\text{s.t.} \quad & x_1 \leq 4, \\
& 2x_2 \leq 12, \\
& 3x_1 + 2x_2 \leq 18, \\
& x_1, \quad x_2 \geq 0.
\end{aligned}
$$

	Doors	Windows			
Profit per batch	$3,000	$5,000			
			Hours		Hours
	Hours used per batch produced		used		available
Plant 1	1	0	2	<=	4
Plant 2	0	2	12	<=	12
Plant 3	3	2	18	<=	18
	Doors	Windows			Total profit
Batches produced	2	6			$36,000

Wyndor Glass Co. (Dual Problem)

- Two products
  - Product 1
    - Glass door w/ aluminum framing
  - Product 2
    - Wood-frame window
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    - Aluminum frames and hardware
  - Plant 2
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	Doors	Windows		
Profit per batch	$3,000	$5,000		
				Hours
	Hours used per batch produced			available
Plant 1	1	0		4
Plant 2	0	2		12
Plant 3	3	2		18

- Decision: production time price
- Objective: to minimize the total price (per week)

*Wyndor Glass Co. (Dual Problem)

Wyndor Glass Co. (Dual Problem)				
	Doors	Windows		
	Hours used per batch produced		Hours available	Unit price
Plant 1	1	0	4	
Plant 2	0	2	12	
Plant 3	3	2	18	
			Total price	
Profit per batch	$3,000	$5,000		

Wyndor Glass Co. (Dual Problem)				
	Doors	Windows		
	Hours used per batch produced		Hours available	Unit price
Plant 1	1	0	4	$0
Plant 2	0	2	12	$1,500
Plant 3	3	2	18	$1,000
Combined price	$3,000	$5,000		
	≥	≥	Total price	
Profit per batch	$3,000	$5,000	$36,000	

*Wyndor Glass Co. (Dual Problem)

- Mathematical formulation
  - π_1 = unit price of production time in plant 1
  - π_2 = unit price of production time in plant 2
  - π_3 = unit price of production time in plant 3
  - w = total price for the production time of plants

Wyndor Glass Co. (Dual Problem)				
	Doors	Windows		
	Hours used per batch produced		Hours available	Unit price
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Plant 2	0	2	12	
Plant 3	3	2	18	
			Total price	
Profit per batch	$3,000	$5,000		

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  - w = total price for the production time of plants

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	Doors	Windows		
	Hours used per batch produced		Hours available	Unit price
Plant 1	1	0	4	
Plant 2	0	2	12	
Plant 3	3	2	18	
			Total price	
Profit per batch	$3,000	$5,000		

$$
\begin{array}{lrrrl}
\min & 4\pi_1 & +12\pi_2 & +18\pi_3 & \\
\text{s.t.} & \pi_1 & & +3\pi_3 & \geq 3, \\
 & & +2\pi_2 & +2\pi_3 & \geq 5, \\
 & \pi_1, & \pi_2, & \pi_3 & \geq 0.
\end{array}
$$

Wyndor Glass Co. (Dual Problem)				
	Doors	Windows		
	Hours used per batch produced		Hours available	Unit price
Plant 1	1	0	4	$0
Plant 2	0	2	12	$1,500
Plant 3	3	2	18	$1,000
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Profit per batch	$3,000	$5,000	$36,000	

Dual: Bounding Problem

Dual: Bounding Problem

- Consider the standard form LP

$$
\begin{array}{lrcl}
\max & 3x_1+8x_2+5x_3+x_4 & & \\
\text{s.t.} & 6x_1+4x_2+2x_3+4x_4 & \leq 12 & (1) \\
 & x_1+2x_2+2x_3-x_4 & \leq 8 & (2) \\
 & x_1,\quad x_2,\quad x_3,\quad x_4 & \geq 0 &
\end{array}
$$

- How to find an upper bound on the optimal value?

*Dual: Bounding Problem

- Consider the standard form LP

$$
\begin{array}{lrcl}
\max & 3x_1+8x_2+5x_3+x_4 & & \\
\text{s.t.} & 6x_1+4x_2+2x_3+4x_4 & \leq 12 & (1) \\
 & x_1+2x_2+2x_3-x_4 & \leq 8 & (2) \\
 & x_1,\quad x_2,\quad x_3,\quad x_4 & \geq 0 &
\end{array}
$$

- If we multiply constraint (1) by 5/2

$$15x_1 + 10x_2 + 5x_3 + 10x_4 \leq (5/2)12 = 30$$

*Dual: Bounding Problem

- Consider the standard form LP

$$
\begin{array}{lrcl}
\max & 3x_1+8x_2+5x_3+x_4 & & \\
\text{s.t.} & 6x_1+4x_2+2x_3+4x_4 & \leq 12 & (1) \\
 & x_1+2x_2+2x_3-x_4 & \leq 8 & (2) \\
 & x_1,\quad x_2,\quad x_3,\quad x_4 & \geq 0 &
\end{array}
$$

- If we multiply constraint (1) by 5/2

$$
\begin{aligned}
z &= 3x_1+8x_2+5x_3+x_4 \\
&\leq 15x_1+10x_2+5x_3+10x_4 \leq (5/2)12 = 30
\end{aligned}
$$

➔ $z^* \leq 30$ (upper bound)

*Dual: Bounding Problem

- Consider the standard form LP

$$
\begin{array}{lrcl}
\max & 3x_1+8x_2+5x_3+x_4 & & \\
\text{s.t.} & 6x_1+4x_2+2x_3+4x_4 & \leq 12 & (1) \\
 & x_1+2x_2+2x_3-x_4 & \leq 8 & (2) \\
 & x_1,\quad x_2,\quad x_3,\quad x_4 & \geq 0 &
\end{array}
$$

- What about constraint (2)?

*Dual: Bounding Problem

- Consider the standard form LP

$$
\begin{array}{lrcl}
\max & 3x_1+8x_2+5x_3+x_4 & & \\
\text{s.t.} & 6x_1+4x_2+2x_3+4x_4 & \leq 12 & (1) \\
 & x_1+2x_2+2x_3-x_4 & \leq 8 & (2) \\
 & x_1,\quad x_2,\quad x_3,\quad x_4 & \geq 0 &
\end{array}
$$

- What about constraint (2)?
  - Negative coefficient of x_4

*Dual: Bounding Problem

- Consider the standard form LP

$$
\begin{array}{lrcl}
\max & 3x_1+8x_2+5x_3 \quad +x_4 & & \\
\text{s.t.} & 6x_1+4x_2+2x_3+4x_4 & \leq 12 & (1) \\
 & x_1+2x_2+2x_3 \quad -x_4 & \leq 8 & (2) \\
 & x_1, \quad x_2, \quad x_3, \quad x_4 & \geq 0 &
\end{array}
$$

- What about (1) + 2 * (2)?

$$
\begin{aligned}
& 8x_1+8x_2+6x_3+2x_4 \\
= & (6x_1+4x_2+2x_3+4x_4)+2(x_1+2x_2+2x_3-x_4) \\
\leq & 12+2(8)=28
\end{aligned}
$$

*Dual: Bounding Problem

- Consider the standard form LP

$$
\begin{array}{lrcl}
\max & 3x_1+8x_2+5x_3 \quad +x_4 & & \\
\text{s.t.} & 6x_1+4x_2+2x_3+4x_4 & \leq 12 & (1) \\
 & x_1+2x_2+2x_3 \quad -x_4 & \leq 8 & (2) \\
 & x_1, \quad x_2, \quad x_3, \quad x_4 & \geq 0 &
\end{array}
$$

- What about (1) + 2 * (2)?

$$
\begin{aligned}
z &= 3x_1 + 8x_2 + 5x_3 + x_4 \\
&\leq 8x_1 + 8x_2 + 6x_3 + 2x_4 \\
&= (6x_1 + 4x_2 + 2x_3 + 4x_4) + 2(x_1 + 2x_2 + 2x_3 - x_4) \\
&\leq 12 + 2(8) = 28
\end{aligned}
$$

➔ $z^* \leq 28$ (upper bound)

*Dual: Bounding Problem

- Consider the standard form LP

$$
\begin{array}{lrcl}
\max & 3x_1+8x_2+5x_3+x_4 & & \\
\text{s.t.} & 6x_1+4x_2+2x_3+4x_4 & \le 12 & (1) \\
 & x_1+2x_2+2x_3-x_4 & \le 8 & (2) \\
 & x_1,\quad x_2,\quad x_3,\quad x_4 & \ge 0 &
\end{array}
$$

- For any nonnegative combination of (1) and (2), i.e., $\pi_1 \ge 0$, $\pi_2 \ge 0$,

$$
\begin{aligned}
z &= 3x_1+8x_2+5x_3+x_4 \\
&\le ?(6\pi_1+\pi_2)x_1+(4\pi_1+2\pi_2)x_2+(2\pi_1+2\pi_2)x_3+(4\pi_1-\pi_2)x_4 \\
&= \pi_1(6x_1+4x_2+2x_3+4x_4)+\pi_2(x_1+2x_2+2x_3-x_4) \\
&\le 12\pi_1+8\pi_2
\end{aligned}
$$

*Dual: Bounding Problem

- Consider the standard form LP

$$
\begin{array}{lrcl}
\max & 3x_1+8x_2+5x_3+x_4 & & \\
\text{s.t.} & 6x_1+4x_2+2x_3+4x_4 & \le 12 & (1) \\
 & x_1+2x_2+2x_3-x_4 & \le 8 & (2) \\
 & x_1,\ x_2,\ x_3,\ x_4 & \ge 0 &
\end{array}
$$

- For any nonnegative combination of (1) and (2), i.e., $\pi_1 \ge 0$, $\pi_2 \ge 0$,

$$
\begin{array}{rl}
6\pi_1 & +\pi_2 \ge 3 \\
4\pi_1 & +2\pi_2 \ge 8 \\
2\pi_1 & +2\pi_2 \ge 5 \\
4\pi_1 & -\pi_2 \ge 1
\end{array}
\quad \Rightarrow \quad
\begin{aligned}
z &= 3x_1+8x_2+5x_3+x_4 \\
&\le (6\pi_1+\pi_2)x_1+(4\pi_1+2\pi_2)x_2+(2\pi_1+2\pi_2)x_3+(4\pi_1-\pi_2)x_4 \\
&= \pi_1(6x_1+4x_2+2x_3+4x_4)+\pi_2(x_1+2x_2+2x_3-x_4) \\
&\le 12\pi_1+8\pi_2
\end{aligned}
$$

  - ➔ $z^* \le 12\pi_1+8\pi_2$ (upper bound)
  - ➔ The smallest upper bound?

*Dual: Bounding Problem

- Consider the standard form LP

$$\begin{array}{lrl} \max & 3x_1+8x_2+5x_3+x_4 & \\ \text{s.t.} & 6x_1+4x_2+2x_3+4x_4 \le 12 & (\pi_1) \\ & x_1+2x_2+2x_3-x_4 \le 8 & (\pi_2) \\ & x_1,\ x_2,\ x_3,\ x_4 \ge 0 & \end{array}$$

$\rightarrow$

$$\begin{array}{ll} \min & 12\pi_1+8\pi_2 \\ \text{s.t.} & 6\pi_1+\pi_2 \ge 3 \\ & 4\pi_1+2\pi_2 \ge 8 \\ & 2\pi_1+2\pi_2 \ge 5 \\ & 4\pi_1-\pi_2 \ge 1 \\ & \pi_1,\ \pi_2 \ge 0 \end{array}$$

Primal Dual

- Any linear program has a corresponding bounding linear program, known as its dual.

Primal-Dual Relationship

Primal-Dual Relationship

- Standard form

$$\boxed{\begin{aligned} \max \quad & z = c^T x \\ \text{s.t.} \quad & Ax \leq b \\ & x \geq 0. \end{aligned}}$$

*Primal-Dual Relationship

- Standard form

$$\begin{aligned} \max \quad & z = c^T x \\ \text{s.t.} \quad & Ax \le b \\ & x \ge 0. \end{aligned}$$

$$A = \begin{array}{c} \\ \\ \\ \\ \end{array} \begin{array}{c} \begin{array}{ccccc} & & A_j & & \end{array} \\ \begin{bmatrix} a_{11} & \cdots & a_{1j} & \cdots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{i1} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & & \vdots & & \vdots \\ a_{m1} & \cdots & a_{mj} & \cdots & a_{mn} \end{bmatrix} \end{array} \begin{array}{c} \\ \\ \\ a_i \\ \\ \end{array}$$

- Let a_i be the ith row and A_j the jth column.

*Primal-Dual Relationship

- Standard form

$$\begin{aligned} \max \quad & z = c^T x \\ \text{s.t.} \quad & Ax \le b \\ & x \ge 0. \end{aligned}$$

$$\begin{array}{lrcl} & & A_2 & \\ \max & 3x_1 + 8x_2 + 5x_3 + x_4 & & \\ \text{s.t.} & 6x_1 + 4x_2 + 2x_3 + 4x_4 & \le 12 & a_1 \\ & x_1 + 2x_2 + 2x_3 - x_4 & \le 8 & \\ & x_1, \quad x_2, \quad x_3, \quad x_4 & \ge 0 & \end{array}$$

- Let a_i be the ith row and A_j the jth column.

*Primal-Dual Relationship

- Standard form

$$\begin{array}{ll} \max & z = c^T x \\ \text{s.t.} & Ax \le b \\ & x \ge 0. \end{array}$$

$$\begin{array}{lrl} & & \quad A_2 \\ \max & 3x_1 + 8x_2 + 5x_3 + x_4 & \\ \text{s.t.} & 6x_1 + 4x_2 + 2x_3 + 4x_4 \le 12 & a_1 \\ & x_1 + 2x_2 + 2x_3 - x_4 \le 8 & \\ & x_1, \quad x_2, \quad x_3, \quad x_4 \ge 0 & \end{array}$$

- Let a_i be the ith row and A_j the jth column.
- For any nonnegative combination, i.e., $\pi \ge 0$

$$\pi_1 a_1^T x + \ldots + \pi_m a_m^T x \le \pi_1 b_1 + \ldots + \pi_m b_m = \pi^T b = b^T \pi$$

$$\begin{array}{l} \pi_1(6x_1 + 4x_2 + 2x_3 + 4x_4) \\ +\pi_2(x_1 + 2x_2 + 2x_3 - x_4) \\ \le 12\pi_1 + 8\pi_2 \end{array}$$

*Primal-Dual Relationship

Standard form

$$\begin{array}{ll} \max & z = c^T x \\ \text{s.t.} & Ax \le b \\ & x \ge 0. \end{array}$$

$$\begin{array}{lrl} & A_2 & \\ \max & 3x_1 + 8x_2 + 5x_3 + x_4 & \\ \text{s.t.} & 6x_1 + 4x_2 + 2x_3 + 4x_4 \le 12 & a_1 \\ & x_1 + 2x_2 + 2x_3 - x_4 \le 8 & \\ & x_1, \quad x_2, \quad x_3, \quad x_4 \ge 0 & \end{array}$$

- Let a_i be the ith row and A_j the jth column.
- For any nonnegative combination, i.e., $\pi \ge 0$

$$\pi_1 a_1^T x + \ldots + \pi_m a_m^T x \le \pi_1 b_1 + \ldots + \pi_m b_m = \pi^T b = b^T \pi$$

$$\begin{aligned} & \pi_1 (6x_1 + 4x_2 + 2x_3 + 4x_4) \\ & + \pi_2 (x_1 + 2x_2 + 2x_3 - x_4) \\ & \le 12\pi_1 + 8\pi_2 \end{aligned}$$

- For $b^T \pi$ to be an upper bound on z^*, requiring

$$\pi^T A_j \ge c_j, \quad j = 1, \ldots, n$$

- or

$$\pi^T A \ge c^T \qquad A^T \pi \ge c$$

$$\begin{array}{lrl} 6\pi_1 & +\pi_2 & \ge 3 \\ 4\pi_1 & +2\pi_2 & \ge 8 \\ 2\pi_1 & +2\pi_2 & \ge 5 \\ 4\pi_1 & -\pi_2 & \ge 1 \end{array}$$

*Primal-Dual Relationship

- Standard form

$$\begin{aligned} \max \quad & z = c^T x \\ \text{s.t.} \quad & Ax \le b \\ & x \ge 0. \end{aligned}$$

- Dual

$$\begin{aligned} \min \quad & w = b^T \pi \\ \text{s.t.} \quad & A^T \pi \ge c \\ & \pi \ge 0. \end{aligned}$$

- Let a_i be the ith row and A_j the jth column.
- For any nonnegative combination, i.e., $\pi \ge 0$

$$\pi_1 a_1^T x + \ldots + \pi_m a_m^T x \le \pi_1 b_1 + \ldots + \pi_m b_m = \pi^T b = b^T \pi$$

- For $b^T \pi$ to be an upper bound on z^*, requiring

$$\pi^T A_j \ge c_j, \quad j = 1, \ldots, n$$

- or

$$\pi^T A \ge c^T \qquad A^T \pi \ge c$$

*Primal-Dual Relationship

- Canonical form

$$\boxed{\begin{aligned} \max \quad & z = c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0. \end{aligned}}$$

- Dual

*Primal-Dual Relationship

- Canonical form

$$\begin{aligned} \max \quad & z = c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0. \end{aligned}$$

$$\begin{aligned} \max \quad & z = c^T x \\ \text{s.t.} \quad & Ax \leq b \\ & -Ax \leq -b \\ & x \geq 0. \end{aligned}$$

- Dual

$$\begin{aligned} \min \quad & w = b^T \pi \\ \text{s.t.} \quad & A^T \pi \geq c \\ & \pi \ u.r.s. \end{aligned}$$

$\pi = \pi^+ - \pi^-$

$$\begin{aligned} \min \quad & w = b^T \pi^+ - b^T \pi^- \\ \text{s.t.} \quad & A^T \pi^+ - A^T \pi^- \geq c \\ & \pi^+, \quad \pi^- \geq 0. \end{aligned}$$

Primal-Dual Relationship

- Primal

$$\begin{aligned} \max \quad & z = c^T x \\ \text{s.t.} \quad & Ax \le b \\ & x \ge 0. \end{aligned}$$

- Dual

$$\begin{aligned} \min \quad & w = b^T \pi \\ \text{s.t.} \quad & A^T \pi \ge c \\ & \pi \ge 0. \end{aligned}$$

Primal		Dual
Constraint	$\Rightarrow$	Variable
Variable	$\Rightarrow$	Constraint
RHS, b	$\Rightarrow$	Objective, c
Objective, c	$\Rightarrow$	RHS, b
Max	$\Rightarrow$	Min
Min	$\Rightarrow$	Max

Primal-Dual Relationship

- Primal

$$\begin{aligned} \max \quad & z = c^T x \\ \text{s.t.} \quad & Ax \le b \\ & x \ge 0. \end{aligned}$$

- Dual

$$\begin{aligned} \min \quad & w = b^T \pi \\ \text{s.t.} \quad & A^T \pi \ge c \\ & \pi \ge 0. \end{aligned}$$

Max		Min	
Constraint	$\le b_i$	≥ 0	Variable
	$= b_i$	u.r.s.	
	$\ge b_i$	≤ 0	
Variable	≥ 0	$\ge c_j$	Constraint
	u.r.s.	$= c_j$	
	≤ 0	$\le c_j$	

Primal-Dual Relationship

- Exercise

$$
\begin{array}{lrl}
\max & -3x_1+2x_2+5x_3 & \\
\text{s.t.} & -x_1+3x_2+4x_3 = 12 & (\pi_1) \\
& 2x_1 \quad -x_2+3x_3 \geq 6 & (\pi_2) \\
& x_3 \leq 4 & (\pi_3) \\
& x_1 \qquad\qquad \geq 0 & \\
& \quad x_2 \qquad \leq 0 & \\
& x_3 \ u.r.s. &
\end{array}
$$

- Dual

*Primal-Dual Relationship

- Exercise

$$
\begin{array}{lrl}
\max & -3x_1+2x_2+5x_3 & \\
\text{s.t.} & -x_1+3x_2+4x_3 = 12 & (\pi_1) \\
& 2x_1 \quad -x_2+3x_3 \geq 6 & (\pi_2) \\
& x_3 \leq 4 & (\pi_3) \\
& x_1 \geq 0 & \\
& x_2 \leq 0 & \\
& x_3 \ u.r.s. &
\end{array}
$$

- Dual

$$
\begin{array}{lrl}
\min & 12\pi_1+6\pi_2+4\pi_3 & \\
\text{s.t.} & -\pi_1+2\pi_2 \geq -3 & \\
& 3\pi_1 \quad -\pi_2 \leq 2 & \\
& 4\pi_1+3\pi_2 \quad +\pi_3 = 5 & \\
& \pi_1 \quad u.r.s. & \\
& \pi_2 \leq 0 & \\
& \pi_3 \geq 0 &
\end{array}
$$

*Primal-Dual Relationship

- Exercise

$$
\begin{array}{lrrrl}
\min & 12\pi_1 & +6\pi_2 & +4\pi_3 & \\
\text{s.t.} & -\pi_1 & +2\pi_2 & & \geq -3 \\
 & 3\pi_1 & -\pi_2 & & \leq 2 \\
 & 4\pi_1 & +3\pi_2 & +\pi_3 & = 5 \\
 & \pi_1 & & & u.r.s. \\
 & & \pi_2 & & \leq 0 \\
 & & & \pi_3 & \geq 0
\end{array}
$$

- Dual

*Primal-Dual Relationship

- Exercise

$$
\begin{array}{lrrrl}
\min & 12\pi_1 & +6\pi_2 & +4\pi_3 & \\
\text{s.t.} & -\pi_1 & +2\pi_2 & & \geq -3 \\
 & 3\pi_1 & -\pi_2 & & \leq 2 \\
 & 4\pi_1 & +3\pi_2 & +\pi_3 & = 5 \\
 & \pi_1 & & & u.r.s. \\
 & & \pi_2 & & \leq 0 \\
 & & & \pi_3 & \geq 0
\end{array}
$$

- Dual

$$
\begin{array}{lrrrrl}
\max & -3x_1 & +2x_2 & +5x_3 & & \\
\text{s.t.} & -x_1 & +3x_2 & +4x_3 & & = 12 \\
 & 2x_1 & -x_2 & +3x_3 & & \geq 6 \\
 & & & x_3 & & \leq 4 \\
 & x_1 & & & & \geq 0 \\
 & & x_2 & & & \leq 0 \\
 & & & x_3 & & u.r.s.
\end{array}
$$

Primal-Dual Relationship

Theorem: For any linear program (P) with the dual (D), the dual of the dual (D) is equivalent to the original linear program (P).

- Proof (standard form)

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\begin{array}{lll}
(P) & \max & z = c^T x \\
 & \text{s.t.} & Ax \le b \\
 & & x \ge 0.
\end{array}
$$

$$
\begin{array}{lll}
(D) & \min & w = b^T \pi \\
 & \text{s.t.} & A^T \pi \ge c \\
 & & \pi \ge 0.
\end{array}
$$

$$
\begin{array}{lll}
(D) & \max & -w = -b^T \pi \\
 & \text{s.t.} & -A^T \pi \le -c \\
 & & \pi \ge 0.
\end{array}
$$

$$
\begin{array}{lll}
(DD) & \min & -z = -c^T x \\
 & \text{s.t.} & -Ax \ge -b \\
 & & x \ge 0.
\end{array}
$$

Duality Theorems

Duality Theorems

Theorem (Weak duality): For any primal LP (P) that is a maximization problem, if x is a feasible solution for the primal and π is a feasible solution for the dual (D), then $c^T x \leq \mathrm{b}^T \pi$.

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$$\pi^T b - c^T x$$

$$(P) \quad \begin{aligned} \max \quad & z = c^T x \\ \text{s.t.} \quad & Ax \leq b \\ & x \geq 0. \end{aligned}$$

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- Proof (standard form)

$$\begin{aligned}\pi^T b - c^T x &= (\pi^T b - \pi^T A x) + (\pi^T A x - c^T x) \\ &= \pi^T (b - Ax) + (\pi^T A - c^T) x \\ &= \pi^T (b - Ax) + (A^T \pi - c)^T x\end{aligned}$$

$$\begin{aligned}(P) \quad \max \quad & z = c^T x \\ \text{s.t.} \quad & Ax \leq b \\ & x \geq 0.\end{aligned}$$

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$$\begin{aligned}(P) \quad \max \quad & z = c^T x \\ \text{s.t.} \quad & Ax \leq b \\ & x \geq 0.\end{aligned}$$

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- Proof (general form)

$$
\begin{aligned}
\max \quad & z = \sum_{j=1}^{n} c_j x_j \\
\text{s.t.} \quad & \sum_{j=1}^{n} a_{ij} x_j \begin{pmatrix} \leq \\ = \\ \geq \end{pmatrix} b_i, \quad i = 1, \ldots, m, \\
& x_j \begin{pmatrix} \geq 0 \\ u.r.s. \\ \leq 0 \end{pmatrix}, \quad j = 1, \ldots, n.
\end{aligned}
$$

Max		Min	
Constraint	$\leq b_i$	≥ 0	Variable
	$= b_i$	u.r.s.	
	$\geq b_i$	≤ 0	
Variable	≥ 0	$\geq c_j$	Constraint
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The difference between the primal and dual objective function values is

$$\begin{aligned}\pi^T b - c^T x &= (\pi^T b - \pi^T A x) + (\pi^T A x - c^T x) \\ &= \pi^T (b - Ax) + (\pi^T A - c^T) x\end{aligned}$$

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&= \pi^T (b - Ax) + (\pi^T A - c^T) x \\
&= \sum_{i=1}^{m} \pi_i (b_i - a_i^T x) + \sum_{j=1}^{n} (\pi^T A_j - c_j) x_j \\
&\geq 0.
\end{aligned}$$

The last inequality is based on the feasibility of the primal and dual solutions and the primal-dual relationship.

$$\begin{aligned}
\max \quad & z = \sum_{j=1}^{n} c_j x_j \\
\text{s.t.} \quad & \sum_{j=1}^{n} a_{ij} x_j \begin{pmatrix} \leq \\ = \\ \geq \end{pmatrix} b_i, \quad i = 1, \ldots, m, \\
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Corollary 1: If x^* is a feasible solution for the primal, π^* is a feasible solution for its dual, and $c^T x^* = \mathrm{b}^T \pi^*$, then x^* and π^* are optimal solutions for the primal and dual, respectively.

Corollary 2: For any linear program
(a) If a primal problem is unbounded then its dual is infeasible.
(b) If the dual of a problem is unbounded then the primal problem is infeasible.

Duality Theorems

Theorem (Strong duality): If a linear program has an optimal solution, then so does its dual and their optimal values are equal.

- Proof

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Theorem (Strong duality): If a linear program has an optimal solution, then so does its dual and their optimal values are equal.

- Proof (canonical form)

$$\begin{aligned} \max \quad & z = c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0. \end{aligned}$$

$$\begin{aligned} \min \quad & w = b^T \pi \\ \text{s.t.} \quad & A^T \pi \geq c \\ & \pi \ u.r.s. \end{aligned}$$

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Proof (canonical form)

If the LP has an optimal solution x^*, then x^* is a BFS.

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If the LP has an optimal solution x^*, then x^* is a BFS.

Assume x^* is nondegenerate. The optimality condition is,

$$\begin{aligned} \max \quad & z = c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0. \end{aligned}$$

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$$\boxed{c_N^T - c_B^T B^{-1} N \leq 0.}$$

For basic variables,

$$\boxed{c_B^T - c_B^T B^{-1} B = 0.}$$

$$\boxed{c^T - c^T B^{-1} A \leq 0.}$$

$$\begin{aligned} \max \quad & z = c^T x \\ \text{s.t.} \quad & Ax = b \\ & x \geq 0. \end{aligned}$$

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Let $\pi^T = c_B^T B^{-1}$, then π is dual feasible, that is,

$$\boxed{c^T - \pi^T A \leq 0} \quad \text{or} \quad \boxed{A^T \pi \geq c}$$

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$$\begin{aligned} \min \quad & w = b^T \pi \\ \text{s.t.} \quad & A^T \pi \geq c \\ & \pi \ u.r.s. \end{aligned}$$

Further, π is dual optimal, that is,

$$\boxed{\pi^T b = c_B^T B^{-1} b = c_B^T x_B^* = c^T x^*.}$$

The primal and dual have the same optimal values.

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- Proof (general form)

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- Proof (general form)

Transformation from general form LP to canonical form LP does not change the optimal value.

Primal-Dual Relationship: Possibilities

- Primal

$$\begin{aligned} \max \quad & z = c^T x \\ \text{s.t.} \quad & Ax \le b \\ & x \ge 0. \end{aligned}$$

- Dual

$$\begin{aligned} \min \quad & w = b^T \pi \\ \text{s.t.} \quad & A^T \pi \ge c \\ & \pi \ge 0. \end{aligned}$$

	Finite optimum	Unbounded	Infeasible
Finite optimum			
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	Finite optimum	Unbounded	Infeasible
Finite optimum	Possible	Impossible	Impossible
Unbounded	Impossible	Impossible	Possible
Infeasible	Impossible		

$$\begin{aligned} \max \quad & x \\ \text{s.t.} \quad & x \ge 1, \\ & x \ge 0. \end{aligned}$$

$$\begin{aligned} \min \quad & \pi \\ \text{s.t.} \quad & \pi \ge 1, \\ & \pi \le 0. \end{aligned}$$

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	Finite optimum	Unbounded	Infeasible
Finite optimum	Possible	Impossible	Impossible
Unbounded	Impossible	Impossible	Possible
Infeasible	Impossible	Possible	Possible

$$\begin{aligned} \max \quad & 2x_1 - x_2 \\ \text{s.t.} \quad & x_1 - x_2 \le 1, \\ & -x_1 + x_2 \le -2, \\ & x_1, \quad x_2 \ge 0. \end{aligned}$$

$$\begin{aligned} \min \quad & \pi_1 - 2\pi_2 \\ \text{s.t.} \quad & \pi_1 - \pi_2 \ge 2, \\ & -\pi_1 + \pi_2 \ge -1, \\ & \pi_1, \quad \pi_2 \ge 0. \end{aligned}$$

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Complementary Slackness

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Theorem (Complementary Slackness): Let x and π be feasible solutions to the primal and the dual problem, respectively. Then x and π are optimal solutions for the two respective problems if and only if:

$$\boxed{\begin{aligned} \pi_i(a_i^T x - b_i) &= 0, \quad \forall i, \\ (c_j - \pi^T A_j)x_j &= 0, \quad \forall j. \end{aligned}}$$

- Proof

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Proof

Define

$$\boxed{\begin{aligned} u_i &= \pi_i(b_i - a_i^T x), \\ v_j &= (\pi^T A_j - c_j)x_j. \end{aligned}}$$

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Based on the primal-dual relationship, the feasibility of x and π imply that

$$\boxed{\begin{aligned} u_i &\geq 0, \quad \forall i \\ v_j &\geq 0, \quad \forall j. \end{aligned}}$$

Max		Min	
Constraint	$\leq b_i$	≥ 0	Variable
	$= b_i$	u.r.s.	
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$$\boxed{\begin{aligned} u_i &\geq 0, \quad \forall i \\ v_j &\geq 0, \quad \forall j. \end{aligned}}$$

In addition, we know that

$$\boxed{\begin{aligned} \pi^T b - c^T x &= \pi^T(b - Ax) + (\pi^T A - c^T)x \\ &= \sum_{i=1}^{m} u_i + \sum_{j=1}^{n} v_j. \end{aligned}}$$

(Necessity) x and π optimal ➔ $c^T x = \mathrm{b}^T \pi$ ➔ $u_i = v_j = 0$, for all i and j.

(Sufficiency) $u_i = v_j = 0$, for all i and j ➔ $c^T x = \mathrm{b}^T \pi$ ➔ x and π are optimal.

Complementary Slackness

- Example: Is $x = (4, 6)$ an optimal solution of the following LP?

$$\begin{array}{lrrl} \max & 9x_1 & +30x_2 & \\ \text{s.t.} & x_1 & & \leq 8 \\ & & 4x_2 & \leq 24 \\ & 3x_1 & +4x_2 & \leq 36 \\ & x_1, & x_2, & \geq 0 \end{array}$$

- Solution

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- Solution
  - The dual:

$$\begin{array}{lrrrl} \min & 8\pi_1 & +24\pi_2 & +36\pi_3 & \\ \text{s.t.} & \pi_1 & & +3\pi_3 & \geq 9 \\ & & 4\pi_2 & +4\pi_3 & \geq 30 \\ & \pi_1, & \pi_2, & \pi_3 & \geq 0 \end{array}$$

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$s_1 = 8 - x_1 = 4 > 0 \Rightarrow \pi_1 = 0.$
Since $x_1 > 0$ and $x_2 > 0$,

$$\begin{array}{rrl} & 3\pi_3 & = 9 \\ 4\pi_2 & +4\pi_3 & = 30 \end{array}$$

So we have $\pi_2 = 9/2$ and $\pi_3 = 3$.
Both $x = (4, 6)$ and $\pi = (0, 9/2, 3)$ are optimal solutions to the primal and dual problems, respectively. $z = c^T x = b^T \pi = 216$.

*Recap

Dual Problem

- Wyndor
  - Perspective: owner vs. buyer

$$\begin{array}{lrl} \max & z = 3x_1 + 5x_2 & \\ \text{s.t.} & x_1 & \le 4, \\ & 2x_2 & \le 12, \\ & 3x_1 + 2x_2 & \le 18, \\ & x_1, \quad x_2 & \ge 0. \end{array}$$

$$\begin{array}{lrl} \min & 4\pi_1 + 12\pi_2 + 18\pi_3 & \\ \text{s.t.} & \pi_1 \qquad\quad + 3\pi_3 & \ge 3, \\ & + 2\pi_2 + 2\pi_3 & \ge 5, \\ & \pi_1, \quad \pi_2, \quad \pi_3 & \ge 0. \end{array}$$

- Bounding problem
  - Any linear program has a corresponding bounding linear program, known as its dual.

$$\begin{array}{lrl} \max & 3x_1 + 8x_2 + 5x_3 + x_4 & \\ \text{s.t.} & 6x_1 + 4x_2 + 2x_3 + 4x_4 \le 12 & (\pi_1) \\ & x_1 + 2x_2 + 2x_3 - x_4 \le 8 & (\pi_2) \\ & x_1, \quad x_2, \quad x_3, \quad x_4 \ge 0 & \end{array}$$

$\Rightarrow$

$$\begin{array}{lrl} \min & 12\pi_1 + 8\pi_2 & \\ \text{s.t.} & 6\pi_1 + \pi_2 & \ge 3 \\ & 4\pi_1 + 2\pi_2 & \ge 8 \\ & 2\pi_1 + 2\pi_2 & \ge 5 \\ & 4\pi_1 - \pi_2 & \ge 1 \\ & \pi_1, \quad \pi_2 & \ge 0 \end{array}$$

Primal-Dual Relationship

- Primal

$$\begin{aligned} \max \quad & z = c^T x \\ \text{s.t.} \quad & Ax \le b \\ & x \ge 0. \end{aligned}$$

- Dual

$$\begin{aligned} \min \quad & w = b^T \pi \\ \text{s.t.} \quad & A^T \pi \ge c \\ & \pi \ge 0. \end{aligned}$$

Primal		Dual
Constraint	$\Rightarrow$	Variable
Variable	$\Rightarrow$	Constraint
RHS, b	$\Rightarrow$	Objective, c
Objective, c	$\Rightarrow$	RHS, b
Max	$\Rightarrow$	Min
Min	$\Rightarrow$	Max

Primal-Dual Relationship

- Primal

$$\begin{aligned} \max \quad & z = c^T x \\ \text{s.t.} \quad & Ax \le b \\ & x \ge 0. \end{aligned}$$

- Dual

$$\begin{aligned} \min \quad & w = b^T \pi \\ \text{s.t.} \quad & A^T \pi \ge c \\ & \pi \ge 0. \end{aligned}$$

Max		Min	
Constraint	$\le b_i$	≥ 0	Variable
	$= b_i$	u.r.s.	
	$\ge b_i$	≤ 0	
Variable	≥ 0	$\ge c_j$	Constraint
	u.r.s.	$= c_j$	
	≤ 0	$\le c_j$	

Theorem: For any linear program (P) with dual (D), the dual of (D) is equivalent to the original linear program (P).

Duality Theorems

Theorem (Weak duality): For any primal LP (P) that is a maximization problem, if x is a feasible solution for the primal and π is a feasible solution for the dual (D), then $c^T x \leq \mathrm{b}^T \pi$.

Corollary 1: If x^* is a feasible solution for the primal, π^* is a feasible solution for its dual, and $c^T x^* = \mathrm{b}^T \pi^*$, then x^* and π^* are optimal solutions for the primal and dual, respectively.

Corollary 2: For any linear program
(a) If a primal problem is unbounded then its dual is infeasible.
(b) If the dual of a problem is unbounded then the primal problem is infeasible.

Theorem (Strong duality): If a linear program has an optimal solution, then so does its dual and their optimal values are equal.

*Primal-Dual Relationship: Possibilities

- Primal

$$\begin{aligned} \max \quad & z = c^T x \\ \text{s.t.} \quad & Ax \le b \\ & x \ge 0. \end{aligned}$$

- Dual

$$\begin{aligned} \min \quad & w = b^T \pi \\ \text{s.t.} \quad & A^T \pi \ge c \\ & \pi \ge 0. \end{aligned}$$

	Finite optimum	Unbounded	Infeasible
Finite optimum	Possible	Impossible	Impossible
Unbounded	Impossible	Impossible	Possible
Infeasible	Impossible	Possible	Possible

Complementary Slackness

Theorem (Complementary Slackness): Let x and π be feasible solutions to the primal and the dual problem, respectively. Then x and π are optimal solutions for the two respective problems if and only if:

$$\boxed{\begin{aligned} \pi_i(a_i^T x - b_i) &= 0, \quad \forall i, \\ (c_j - \pi^T A_j)x_j &= 0, \quad \forall j. \end{aligned}}$$

Q & A